

ME5412 Robotics for Healthcare

Mobility aids

Cecilia Laschi

Department of Mechanical Engineering

cecilia.laschi@nus.edu.sg

Outline of the Lecture

- Mobility and mobility limitation
- Wheelchair basics and classification
- Manual wheelchairs
- Power-assist mechanisms
- Smart wheelchairs

Mobility

- **Mobility** is fundamental to each individual's quality of life and is necessary for functioning in each of the performance areas:
 - self-care,
 - work or school,
 - play or leisure.
- Mobility is a **foundational** activity for many other skills and activities
 - It enables a person to move to or about a location where the activity will be completed, whether that is within a building or in the community and beyond.
- In addition to the functional gain of increased independence in mobility, other goals such as positive self-image, social interaction, and health maintenance are achieved.

Personal Mobility

- **Personal mobility** is the ability to move oneself from one place to another, within a building, around and outside a building, or between buildings.
- Personal mobility is distinguished from mobility achieved through the use of any form of transportation, private or public
- The World Health Organization's International Classification of Functioning, Disability and Health (ICF) identifies mobility as an **activity**, rather than an impairment of a body structure or function (WHO, 2001)

Mobility Limitations

Significant increase in the number of individuals using mobility aids, across many countries, related to 3 trends:

1. the increasing proportion of older adults in many countries
 - The proportion of individuals reporting a disability increases substantially with age
 - Age-related physical changes such as arthritis result in mobility impairments
2. the rising rates of obesity
 - Mobility devices specifically designed to support the increased size and weight of these individuals
 - **Bariatric chairs** for these individuals whose mobility is impaired by obesity and related chronic diseases
3. accessibility legislation
 - Mandating construction of physical environments that are accessible to individuals who use a wheelchair
 - Defining funding mechanisms to support the acquisition of mobility devices.

Causes of Mobility Limitations

- Neurological disorders
 - Stroke, cerebral palsy, Guillain-Barré syndrome, Huntington's chorea, traumatic brain injury, muscular dystrophy, Parkinson's disease, poliomyelitis, spinal cord injury, spina bifida, and multiple sclerosis
 - Symptoms commonly seen in these neurological disorders are muscle weakness or paralysis, sensory deficits, and abnormal muscle tone
- Musculoskeletal disorders
 - Orthopedic and rheumatological conditions, ankylosing spondylitis, osteogenesis imperfecta, osteoporosis (loss of bone density), Paget's disease, and scoliosis, lower extremity amputation, acquired or congenital
 - Symptoms commonly seen in individuals with arthritis include painful, swollen, and stiff joints (particularly in hand and wrist), muscle weakness, muscle wasting around the affected joints, complaints of feeling fatigued, and, in later stages, joint contractures resulting in range-of-motion limitations
- Cognitive disorders
 - Concomitant disorders that affect an individual's cognitive functioning and ability to learn, such as Alzheimer's disease and cognitive impairment
- Chronic disorders like diabetes, cardiorespiratory conditions, and obesity
 - Fatigue or restrictions related to energy expenditure
 - Amputations resulting from complications due to diabetes

Bear in mind that not all individuals with a given diagnosis experience a similar impairment in mobility

Can you name a mobility aid?

 0

Nobody has responded yet.

Hang tight! Responses are coming in.

Mobility Aids

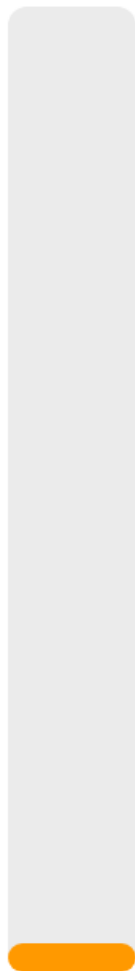
- Limitations to functional mobility can be
 - **augmented** with low-tech AT such as canes, walkers, or crutches
 - **replaced** by wheeled mobility systems of various types
- Lack of access to mobility devices may be a limitation to full participation in society



Can robotic technologies help develop mobility aids?

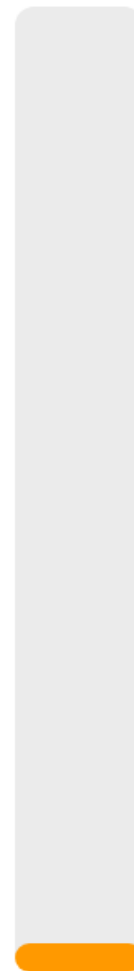
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Yes

0%



No

Mobility aids

- Limitations to functional mobility can be
 - **augmented** with low-tech AT such as canes, walkers, or crutches
 - **replaced** by wheeled mobility systems of various types
- Lack of access to mobility devices may be a limitation to full participation in society
- **Robotics technologies** can contribute to enhance an individual's mobility with smart power wheelchair systems.



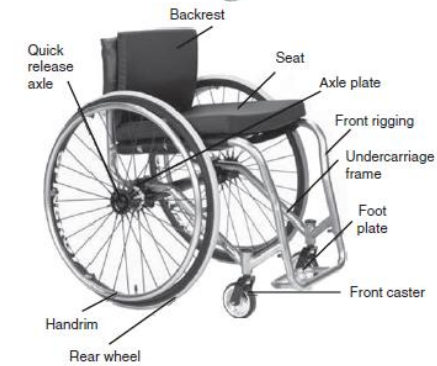
Scope of Mobility Limitations

- 
- **Full ambulator:** no mobility impairment
 - **Marginal ambulator:**
 - can walk short distances
 - may need wheelchair at times, particularly outside the home
 - **Manual wheelchair user:**
 - has some method of propelling a manual wheelchair, whether it is with (1) both upper extremities, (2) both lower extremities, or (3) one upper and one lower extremity , or a caregiver
 - **Marginal manual wheelchair user:**
 - manual wheelchair is used part of the time and powered wheelchair part of the time
 - **Totally/severely mobility-impaired user:**
 - unable to propel self independently in a manual wheelchair
 - dependent mobility system (propelled by an attendant) or
 - Independent powered mobility system

Mobility Aid Classification

Three broad categories of wheeled mobility systems:

- **Dependent mobility systems** are propelled by an attendant and include strollers and transport chairs, as well as manual wheelchairs
A dependent mobility system is chosen when (1) the individual is not at all capable of independently propelling a wheelchair or (2) a secondary system is needed that is lightweight and easily transported.
- An **independent manual mobility system** is for those individuals who have the ability to propel a wheelchair manually.
These bases have two smaller front wheels and two larger wheels in the back (drive wheels) that allow the user to propel independently.
- **Independent power mobility systems** are required when the user has difficulty propelling a manual wheelchair.
These are **power wheelchairs** that are driven by the user.



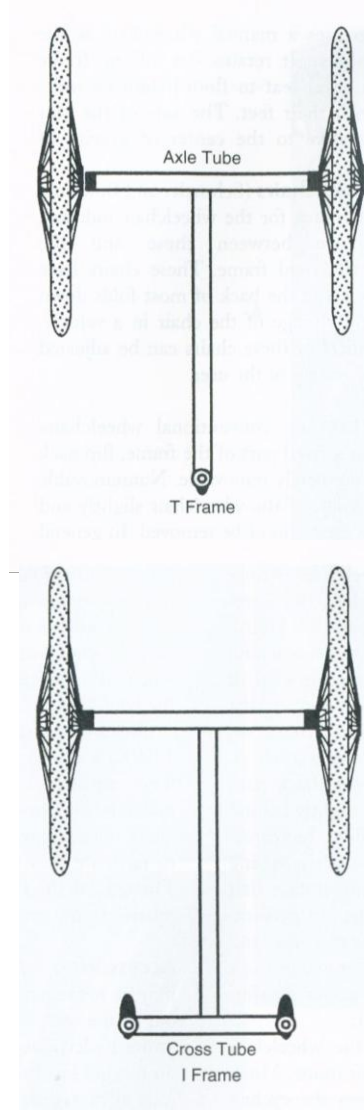


Wheelchair Main Components

Supporting structure

Supporting Structure

- Frame:
 - type of frame: rigid or folding
 - position of the axle of the rear wheels: adjustable or not
 - material used to construct the wheelchair frame: steel, aluminum, other
 - frame style: box, cantilever, T or I frame
- Attachments (armrests, footrests, ...)
- (Optional) Specialized seating and positioning

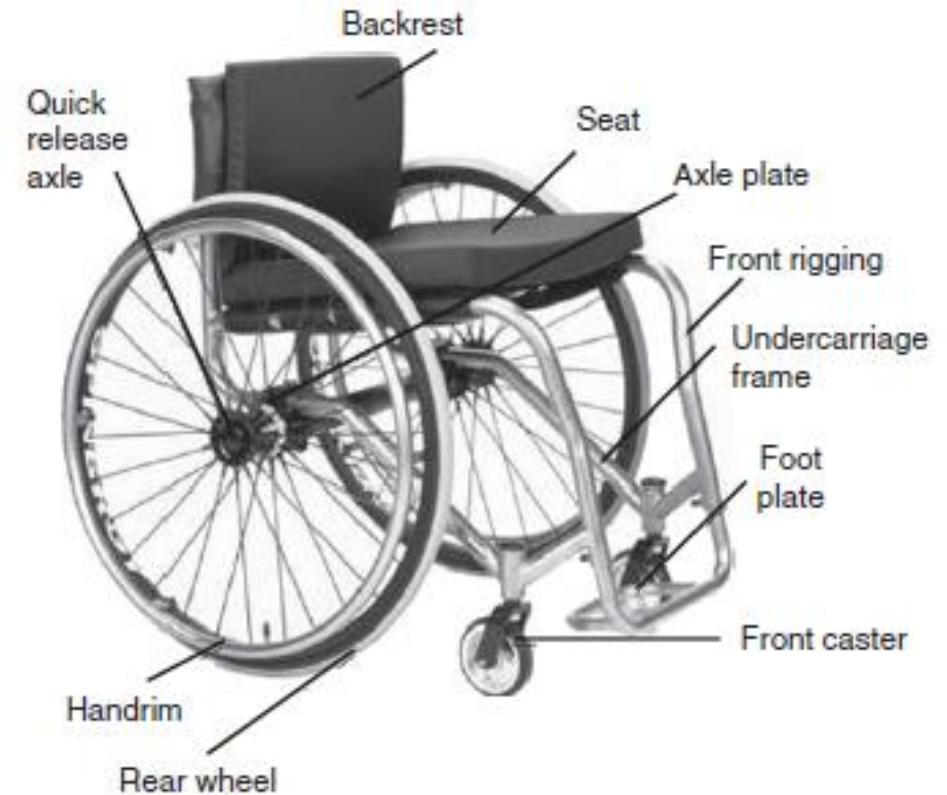


Wheelchair Anatomy

Folding manual wheelchair



Rigid frame manual wheelchair



Tilting and Reclinable Supporting Structures

A. with **tilt** feature

Ability to rotate a specific seating position around a fixed axis



B. with **recline** feature

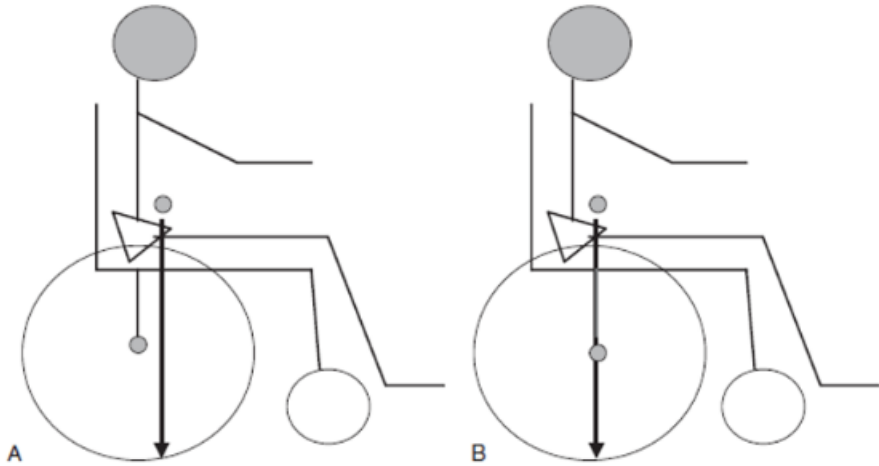
Changing the seat-to-back angle, resulting in greater than 90 degrees



Wheelchair Classification

- Standard wheelchairs
 - Folding steel chairs
 - Fixed axle of rear wheels
 - Heavy and not useful for long-term use
 - Short-term use, e.g., in airports, malls, ...
- Lightweight wheelchairs
 - Steel/aluminium alloy frames
- Ultra-lightweight wheelchairs
 - Axle of rear wheels adjustable to the user's centre of gravity
 - Useful for an individual who uses a manual wheelchair as the primary means of mobility
- Rigid sport ultra-lightweight wheelchairs
 - Rigid frame
 - Titanium/carbon fiber composites
 - Axle of rear wheels adjustable to the user's centre of gravity

Relationship of the user's centre of gravity to the axle of rear wheels



A is more manouvrable



B is more stable



A is more stable



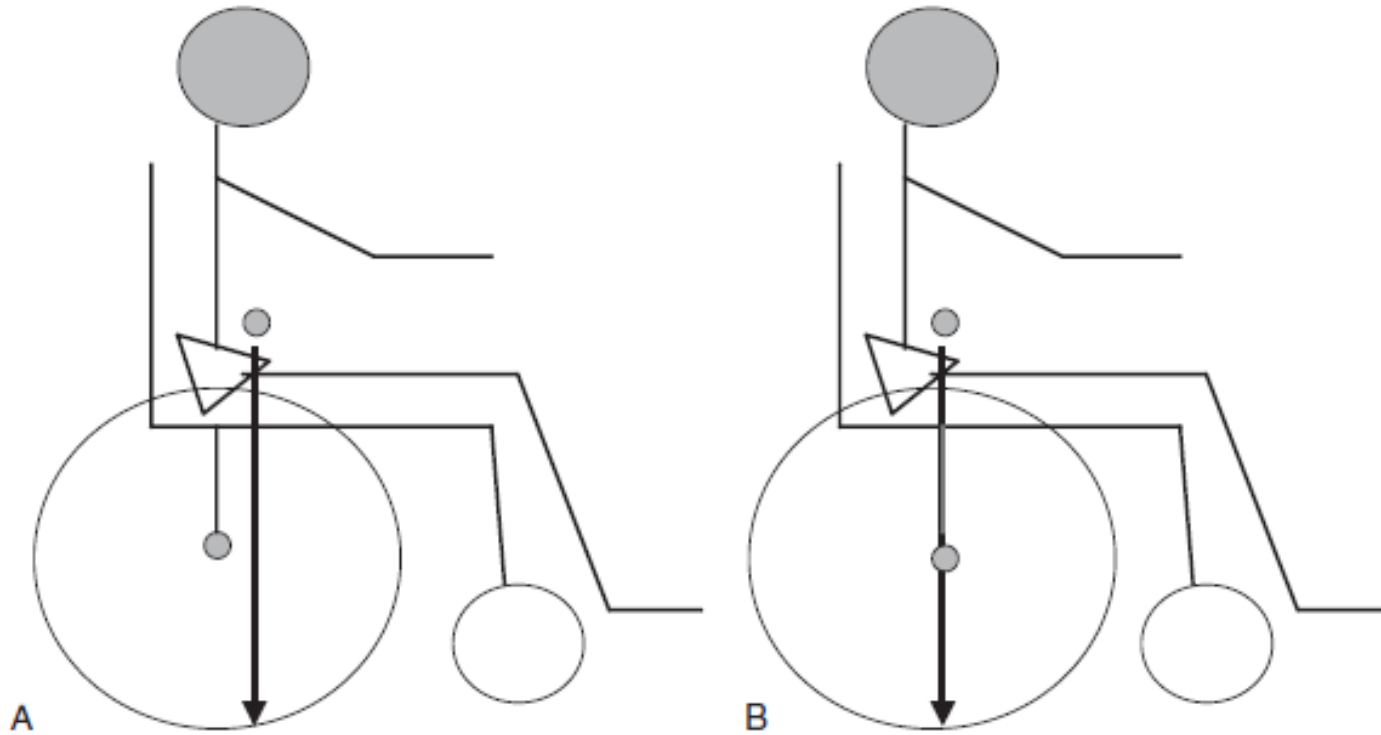
B is more manouvrable



A and B have same stability and manouvrability



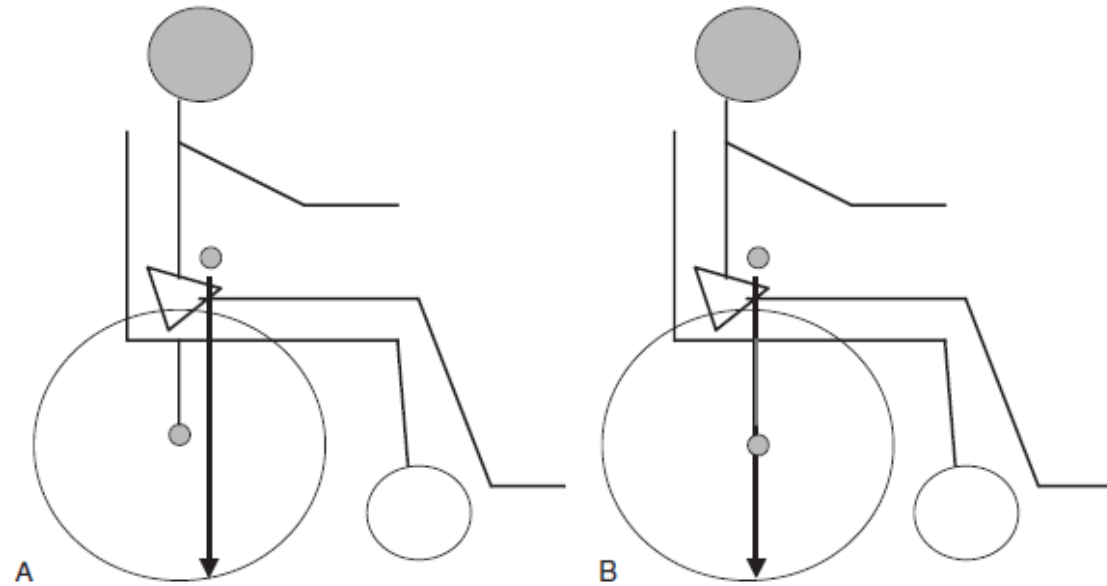
Relationship of the User's Centre of Gravity to the Axle of Rear Wheels



- A. When the user is seated with his center of gravity ahead of the axle of the wheelchair, the chair is **more stable**.
- B. When the user is seated with his center of gravity directly above the axle of the wheelchair, the chair is **more maneuverable**.

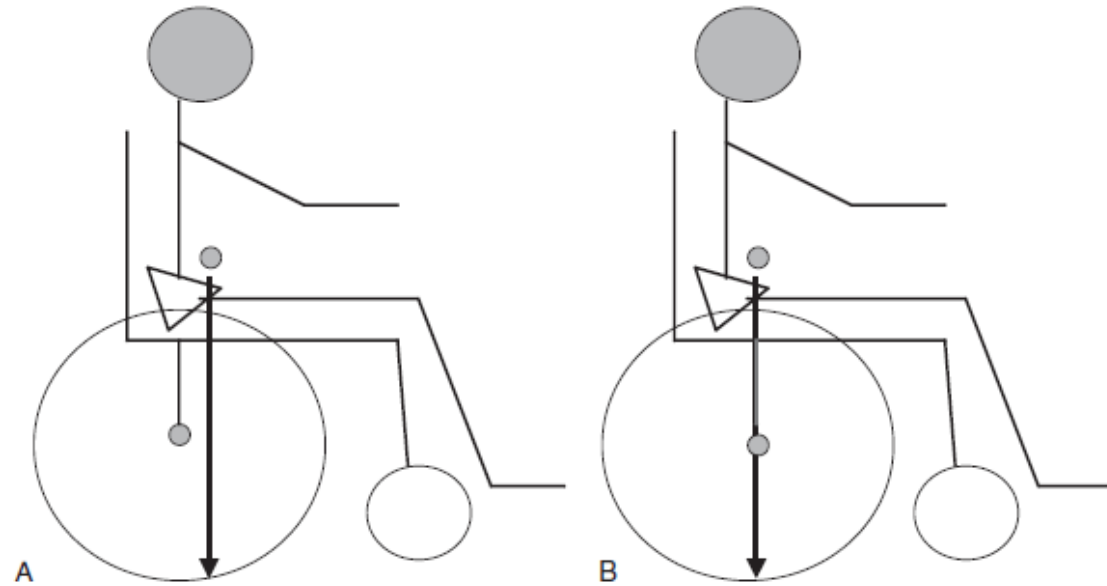
Relationship of the User's Centre of Gravity to the Axle of Rear Wheels

A or B?



Relationship of the User's Centre of Gravity to the Axle of Rear Wheels

A or B?



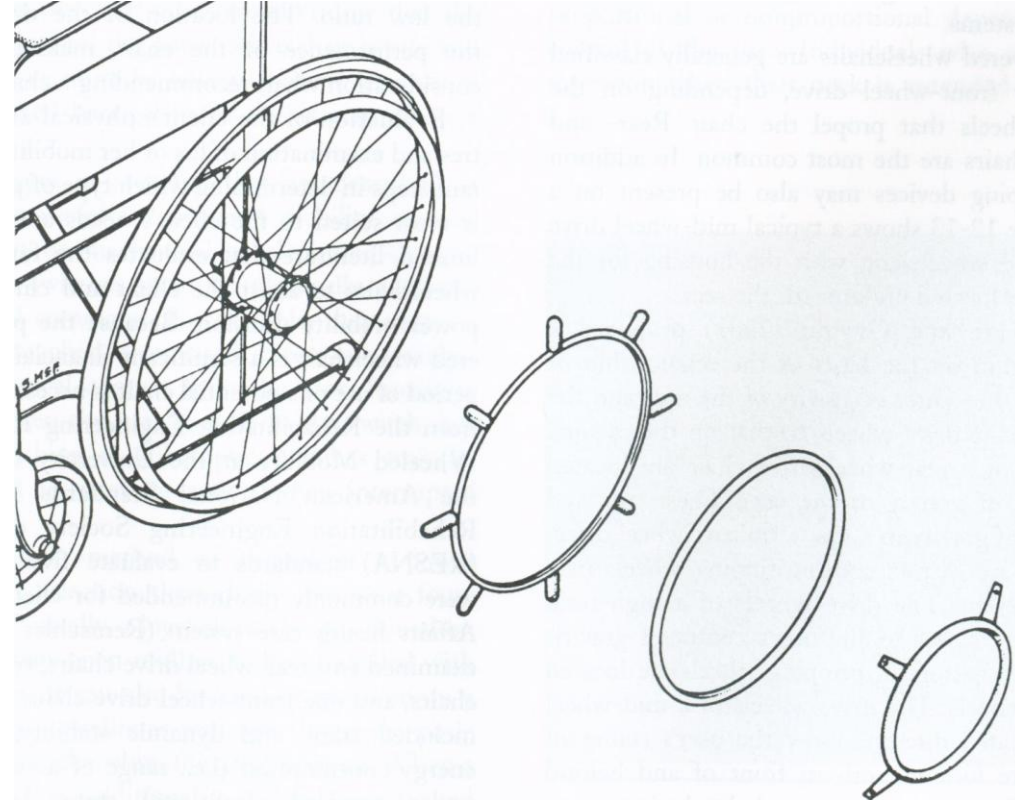


Wheelchair Main Components

Propelling structure

Propelling Structure of Manual Wheelchairs

Hand rims



Power-assist Mechanism

- Manual wheelchair use may cause shoulder injuries
- Power-assist wheels amplify the user's pushing force



Power-assist Mechanism

$$\tau_m(t) = k_m i(t)$$

$\tau_m(t)$: motor torque

k_m : motor torque constant

$i(t)$: motor current

$$\text{output torque} = k_g \tau_m(t)$$

k_g : gear ratio

$$u = k_w k_g \tau_m(t) = k_w k_g k_m i(t)$$

u : control signal

k_w : wheel ratio

$$i(t) \xrightarrow{k_m} \tau_m(t) \xrightarrow{k_g} \text{gear output} \xrightarrow{k_w} u$$

Power-assist Mechanism – Hand Rim Input

$$u = k_w k_p \tau_p(t)$$

u : control signal

k_w : wheel ratio

k_p : power-assist ratio

$\tau_p(t)$: hand rim torque

$$u = 0 \quad | \tau_p(t) | < \varepsilon$$

$$u = k_w k_p \tau_p(t) \quad | \tau_p(t) | \geq \varepsilon$$

$$k_w k_g k_m i(t) = k_w k_g \tau_m(t) = u = k_w k_p \tau_p(t)$$

Power Wheelchairs

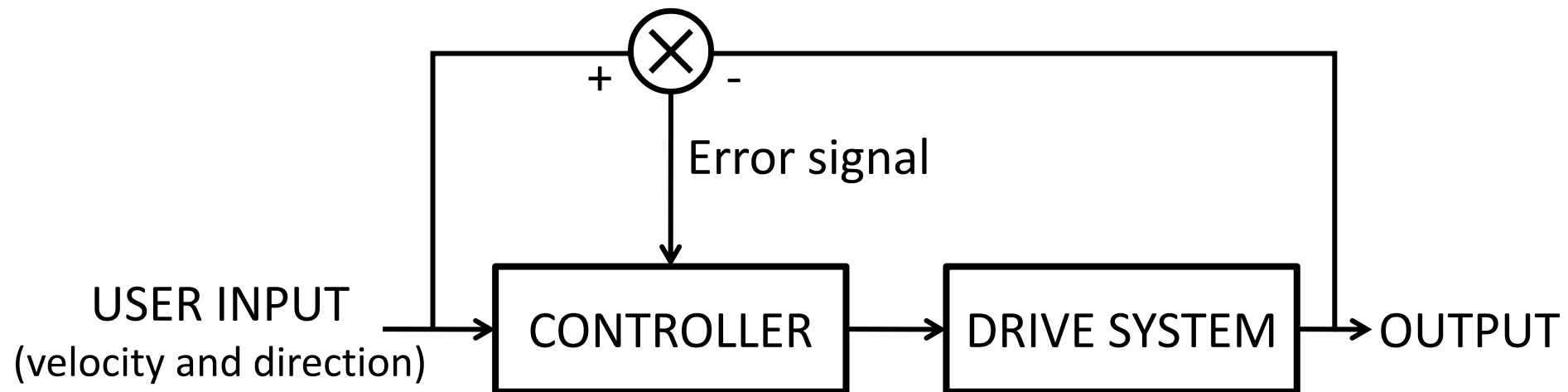
- Power drive to the wheels
- Power is delivered to one pair of wheels (i.e., the drive wheels) with additional pairs of wheels providing stability
- Classified as:
 - Rear-wheel drive
 - Mid-wheel drive
 - Front-wheel drive



Power Wheelchair Control Interfaces

- **Proportional control** with a continuous joystick
 - 360-degree directionality, which means that the chair moves in whichever direction the continuous joystick is displaced.
 - The greater the joystick displacement with respect to its rest position, the faster the chair moves
- **Nonproportional control** with a discrete joystick or switches
 - Movement is activated one direction at a time
 - If the user wishes to change direction, he/she must release the control in one direction and activate it in the direction of the change
 - This type of control is typically used when a user needs to drive the chair using switches.

Power Wheelchair Control



How to make a power wheelchair robotic?



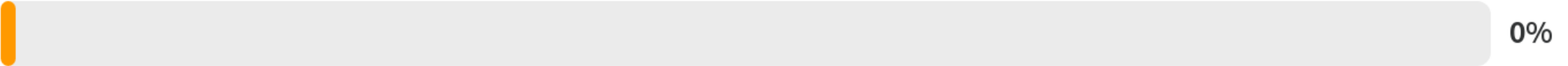
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Hang tight! Responses are coming in.

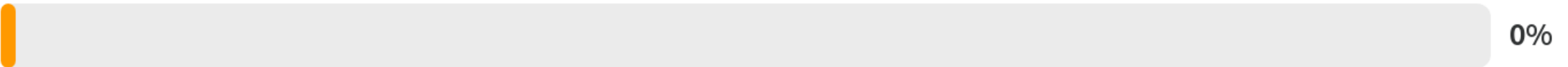
How to make a power wheelchair robotic?



A standard power wheelchair to which a computer and a collection of sensors have been added



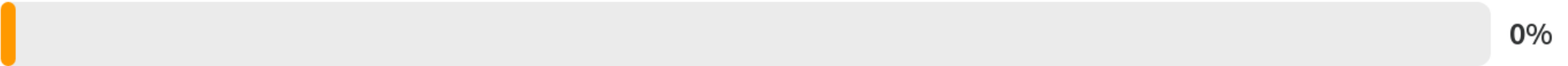
A mobile robot base to which a seat has been attached



Both of the above possible



None of the above: newly designed system



Smart Wheelchairs

- Two approaches:
 - a standard power wheelchair to which a computer and a collection of sensors have been added
 - a mobile robot base to which a seat has been attached
- Useful for wheelchair users who have
 - low vision or severely restricted visual field,
 - motor impairments such as excessive tone or tremor, or
 - cognitive impairments that limit the ability to navigate a wheelchair safely

What robotic functions in a smart wheelchair?

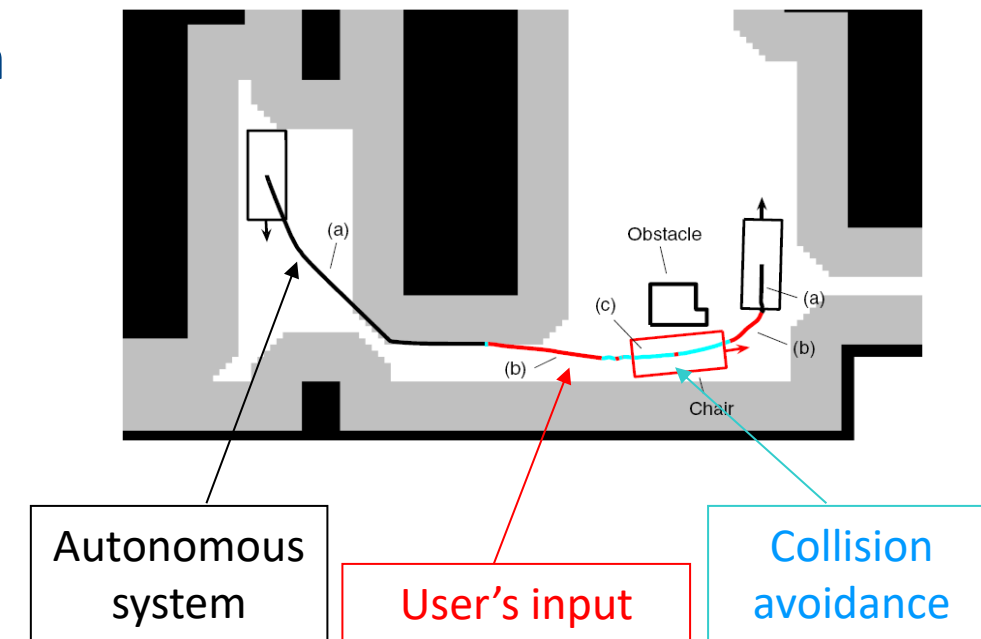


Nobody has responded yet.

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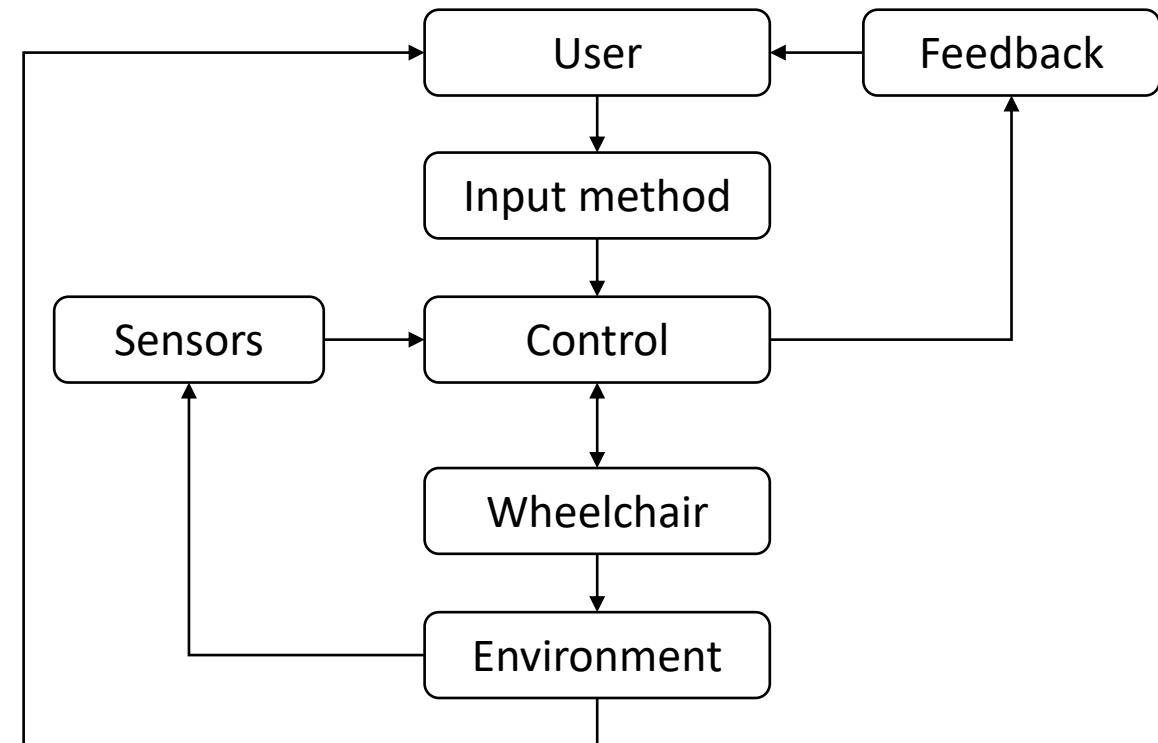
Smart Wheelchairs

- Two functions:
 - **collision avoidance**: sensors detect an obstacle in the path of the wheelchair and will slow or stop the chair or provide some form of warning if the user does not perform a maneuver to avoid it
 - **navigation**, either along a path or through obstacles: the sensors guide the chair along a pathway that is programmed into the chair's control system or that follows an environmental path, or the sensors guide the chair through a feature such as a doorway.



Smart Wheelchair Components

- **Sensors** for identifying features in the environment
- **Input methods** for the user to provide commands to the system
- Methods for the system to provide **feedback** to the user
- **Control** algorithm that coordinates inputs from the user, the sensors and the PWC to determine the system behaviour



What sensors in a smart wheelchair?



Nobody has responded yet.

Hang tight! Responses are coming in.

Smart Wheelchair Sensors

A variety of sensors have been used in smart wheelchairs, but no sensor (or combination of sensors) has been identified that is sufficiently accurate, inexpensive, small, lightweight, and impervious to environmental conditions (e.g., lighting, precipitation, temperature) and requires little power.

TABLE 5.1 Summary of Sensors Used by Smart Wheelchairs

Sensor	Capability	Advantages	Disadvantages
Ultrasonic (sonar)	Measures distance to obstacles by emitting sound waves	• Small • Low power • Low cost	• Sensitive to obstacle properties (such as sound-absorbing material) • Cannot detect very small/thin or concave obstacles • Cross-talk issues with other sounds or multiple ultrasonic sensors • Cannot detect drop-offs unless sensor is pointed to the floor • Limited coverage
Infrared (IR)	Measures distance to obstacles by emitting light; can also detect transitions from light to dark lines	• Small • Low power • Low cost	• False positives in natural light (due to IR interference) • Sensitive to flooring materials (issues with dark, light-absorbent, transparent, and refractive surfaces) • Limited coverage
Laser range finder	Provides 180° scan of obstacles (distances) in the environment	• High precision • Good for longer ranges	• High cost • High power • Possible eye safety issues • Only detects objects at the height of the laser range finder
Stereovision camera	Captures color or black-and-white image and depth images of the environment	• Low power • Medium cost • Can be used for high-level scene understanding (e.g., object and location recognition) and head/eye tracking of users	• Cannot detect textureless objects • Cannot perform in poorly lit conditions • Challenges posed by reflective and transparent surfaces

Smart Wheelchair Sensors (contd.)

TABLE 5.1 Summary of Sensors Used by Smart Wheelchairs

Sensor	Capability	Advantages	Disadvantages
Structured light (e.g., Microsoft Kinect)	Captures color or black-and-white image and depth images (using projected IR) of the environment	• Low power • Low cost • Can be used for high-level scene understanding (e.g., object and location recognition) and head/eye tracking of users	• Challenges posed by outdoor environments (due to IR interference) • Sees through transparent surfaces (i.e., cannot be used to avoid collisions with glass doors) • Can demand a lot of processing power
Bump (tactile sensor)	Can detect if contact is made (force is exerted) by an object	• Low cost • Low power	• Requires contact (safety issues) • Increasing coverage (e.g., by using bumper skirt-like sensors) can lead to increased form factor and bulkiness
Wheelchair encoders	Measures number and speed of wheel rotations	• Can be used to easily gather information on distances traveled and speed	• Can be expensive to retrofit on wheelchairs • Wheel slippage can lead to errors
Inertial measurement units (IMUs)	Contain accelerometers, gyroscopes, and optional magnetometers that can be used to determine velocity, orientation, and position, respectively	• Low cost • Low power	• Prone to accumulated errors when used for longer distances • Higher-accuracy IMUs can be expensive
Global positioning system (GPS)	Provides location and time information	• Can give accurate location when GPS satellites are in an unobstructed line of sight • Low cost	• Does not work well indoors

Smart Wheelchair Input Methods

Smart wheelchairs have been used to explore a variety of alternatives to the more “traditional” input methods associated with power wheelchairs (e.g., joystick, pneumatic switches).

TABLE 5.2 Alternative Input Methods Used in Smart Wheelchairs

Input	Capability	Advantages	Disadvantages
Brain-computer interface	Sensors may be worn on the head (e.g., electroencephalography, EEG) or implanted in the skull (e.g., electrocorticography, ECG).	• Requires no physical movement	• Requires significant computational power • Expensive • Limited signals available for control • Sensors must be placed on or inside the skull
Eye gaze	Cameras pointed at the user’s face track where the user is looking to determine where the PWC should steer.	• Requires limited physical movement • No sensors touch the user’s body	• Requires significant computational power • Distinguishing between “looking to drive” and “looking to see” is challenging • Expensive
Voice recognition	Speech recognition software interprets steering commands from user.	• Low cost • No sensors touch the user’s body	• Requires ability to speak clearly and consistently • Voice has limited bandwidth

Smart Wheelchair Feedback Modalities

- Smart wheelchairs can provide feedback to the driver through various modalities and interfaces:
 - **speakers** can be used to provide audio feedback
 - small light-emitting diodes (**LEDs**) can be used to offer visual feedback (such as a flashing red light if the driver gets too close to an object)
 - **haptic** feedback can be provided by using small vibrations to warn the driver about obstacles
 - **touch screen displays** can be used both to gather information from the driver (e.g., desired destination) and to display information (e.g., scheduled activities, way-finding assistance, virtual environments, etc.).

Smart Wheelchair Feedback Modalities (cntd.)

TABLE 5.3 Summary of Feedback Modalities Used by Smart Wheelchairs

Modality	Implementation	Advantages	Disadvantages
Auditory	Feedback can be provided through intermittent sound (e.g., warning sirens), continuous sound (e.g., tone indicating distance to the nearest obstacle), or spoken words.	• Does not distract visual attention • Low cost	• May be washed out by ambient noise • Nonspoken feedback may be misinterpreted • Only available for a short period of time
Visual	Feedback can be provided through something as simple as individual lights or as complex as a “dashboard” implemented on a computer screen.	• Not affected by ambient noise • Remains available to user • Low cost	• Can distract visual attention
Haptic	Feedback can be provided through pressure applied to a joystick.	• Possible to give very precise feedback about the environment around the PWC • Does not distract visual attention	• Only works if the driver uses a joystick to operate the PWC • Expensive

Smart Wheelchair Control

- A variety of approaches to implement control software for smart wheelchairs, based on the functions supported by the smart wheelchair and the sensors it uses
- Two main control approaches:
 - Autonomous control
 - Shared control, or semiautonomous control

Smart Wheelchair Control

Autonomous control system:

- The onboard computer assumes full control of the chair
- Sensors are used to guide the chair

Name a technique that can be used for autonomous navigation of a smart wheelchair



Nobody has responded yet.

Hang tight! Responses are coming in.

Smart Wheelchair Control

Autonomous control system:

- The onboard computer assumes full control of the chair
- Sensors are used to guide the chair

Shared control (or semiautonomous) system:

- The control of the wheelchair is shared between the user and the onboard computer
- It consists of combining or switching between input from the driver and the intelligent system
- This method of control offers the driver higher levels of control while ensuring safety
- It may be more acceptable to users because they retain some control and do not experience the sensation of the chair moving without their guidance
- However, the exact mechanism of semiautonomous control that is both effective and acceptable to users is still under investigation

Possible smart wheelchair shared control



The user drives, the smart wheelchair ensures that obstacles are avoided

The user delegates control to the smart wheelchair when going through doors or along hallways

The user drives and the smart wheelchair gives warnings on close obstacles

The user sets a destination and/or waypoints and the smart wheelchair navigates to them autonomously

The user sets a direction and the smart wheelchair follows the direction avoiding obstacles, until a new direction is set

Smart Wheelchairs – Discussion

- As might be anticipated, the cost of these systems is extremely high
- A significant technical obstacle to commercialization is the cost/accuracy trade-off that must be made with existing sensors
- Available plug and play technology is likely to reduce this cost in the near future
- There is reason for optimism, however, that the significant research in self-driving cars will lead to affordable technology that can also be used in smart wheelchairs
- In 2014, the technology used in one of Google's self-driving cars cost approximately \$320,000, but industry reports predicted that the additional price a consumer will pay for self-driving car technology will be approximately \$5,000 by 2030 (IHS Automotive 2014)
- Since approximately two-thirds of the cost of the technology underlying self-driving cars is actually devoted to GPS technology, the additional price of technology for a smart wheelchair might be as low as \$2,000

Smart Wheelchairs – Discussion

- Smart technology for wheelchairs is not available on the commercial market*, existing only in research settings
- Smart wheelchairs are at TRL4**: low-fidelity systems used to demonstrate basic functionality in controlled environments
- However, research continues to make this technology more accessible, reliable, and safe.

** validated in a laboratory environment

*see next slide

Smile Smart System

- Power wheelchairs provided with the Smile Smart System (SSS) features, including **line following** capacity and **anti-collision sensors**
- Smile Smart System controls provide a powerchair with safety, as it facilitates autonomous movement regardless of vision levels or muscle patterns.



Smile Smart System

Key System Elements



Controls

Allow numerous input options with simple selection and no need for computer programming.



Bumper

Protects the chair and the driver. Over time the bumper can be removed as driving skills improve.



Sensors

Detect objects in front of the powerchair to avoid collisions and reduce damage.



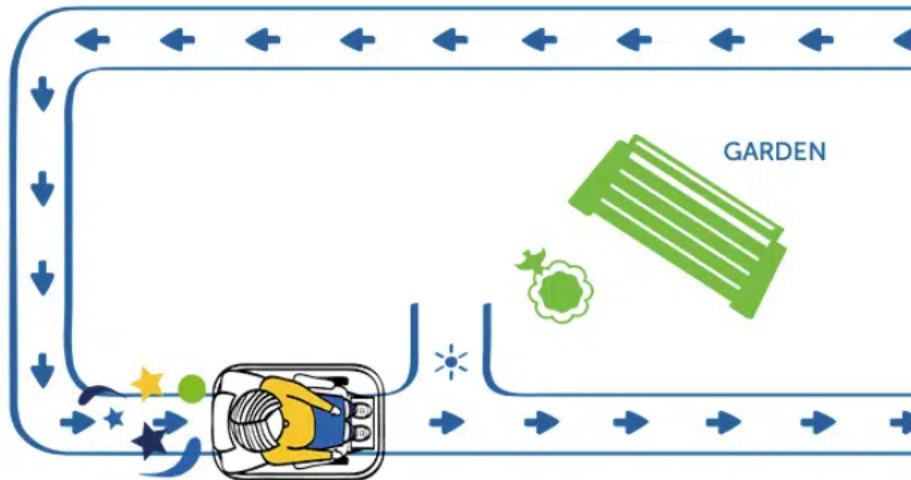
Tape

Indoor and outdoor, permanent or temporary guidance system to steer the powerchair and avoid collisions.



Smile Smart System

Line following



Explore



Line following can be used as a safe pathway from which to develop switch access skills and joystick use over time, gently adapting settings as personal abilities evolve.

Smile Smart System

Anti-collision sensors



Motivate



Anti-collision sensors and bumpers are a brilliant way to prevent damage. Fewer repairs means reduced costs, limited interruptions and allows more freedom & smiles.

Smile Smart System



Personal Drive Prescription

Combining all the available elements of the Smile Smart System, our team are able to go that step further by creating unique mounting, switching, controls and accessory formations.

The extensive range of adaptive switches, control devices and mounting solutions at Smile Smart Technology enable users to complete a wide range of daily tasks with ease.

Many of the switches and control solutions have been developed in-house by Smile Smart Technology. Our holistic approach to mounting along with a commitment to good aesthetics help us provide user-friendly solutions, all based on the feedback from users and support teams.

Summary

- Personal mobility is a key activity for ADLs
- Mobility limitations are growing, to a variety of health conditions, and restrict an individual's participation
- Wheelchairs are the most common mobility aid, replacing (not augmenting) the individual mobility
- Wheelchairs range from manual to power wheelchair
- Robotic technologies can contribute to smart power wheelchair, with obstacle avoidance and navigation capability, with autonomous or shared control